

GEETHIKA ELAPROLU

geethikasiri666@gmail.com | +1 (930) 333-2205 | Bloomington, IN | Open to Relocation | [linkedin.com/in/geethika-elaprolu](https://www.linkedin.com/in/geethika-elaprolu)

EDUCATION

Master's - Data Science, (Indiana University Bloomington, USA) - GPA (3.7/4.0) **Aug 2023 - May 2025**
Coursework: Statistics, Advanced Database Concepts, Database Design, Generative AI, Data Visualization, Applied Machine Learning

Bachelor's - Electronics & Communication Engineering, (KL University, India) - CGPA (9.2/10.0) **Aug 2018 - Apr 2022**
Specialization: Data Computing (Business Analytics, Pattern Recognition, Bigdata Analytics, Data Modeling, Data Management)

PROFESSIONAL EXPERIENCE

Data Engineer Intern **May 2024 – May 2025**
Indiana University

- Performed advanced data analysis on over 1TB+ structured datasets (Parquet) in Azure Data Lake Storage (ADLS), applying query tuning techniques such as caching and distributed execution in Synapse SQL, reducing report generation time by 30%.
- Automated **Tableau** dashboards by scheduling PySpark jobs and integrating SQL queries with live database connections, enabling real-time insights across 6+ departments and accelerating data-driven decision making.
- Built a predictive model using Logistic Regression and QDA to forecast course registrations, improving registration accuracy by 22% over historical trends and optimizing scheduling for 10K+ students.
- Enhanced data pipelines by implementing SCD (Type 1 & 2) and Change Data Capture (CDC) in Azure Data Factory, increasing incremental data accuracy by 45% and minimizing redundant storage by 35%.

Data Analyst **May 2022 - Jul 2023**
Deloitte

- Analyzed large datasets (100K+ records) using advanced Excel functions and SQL to automate reports and conduct ad-hoc analysis, reducing data processing time by 30% and enabling data-driven decisions across business units.
- Structured scalable data models in Power BI by applying SQL techniques such as indexing, partitioning, and schema design, enabling predictive analytics that contributed to a 12% revenue increase across 4+ business channels.
- Created 15+ Power BI dashboards using **DAX measures**, integrating data from Excel and Dataverse to deliver insights and apply **data storytelling** for 5+ stakeholders, improving data accessibility by 60% while ensuring data governance, validation, and PII security through Role-Based Access Controls (RBAC).
- Forecasted audit Key Performance Indicators (**KPIs**) and financial risk with a Random Forest model (75% accuracy), cutting audit cycles by 30% and enhancing risk insight by 25% through automated metrics reporting.
- Identified root causes of slow-running SQL queries and implemented indexing, partitioning, and materialized views, reducing execution time from 35 to under 15 minutes for 3,000+ daily queries.
- Modeled Teradata data warehouses using ERwin by creating logical (LDM) and physical (PDM) database models for 120M+ records, documenting master data management (MDM) rules to ensure data integrity across 10+ sources in the OLAP infrastructure.
- Collaborated with cross-functional BI audit data teams to develop 10+ impactful data visualizations, effectively communicating complex analytical insights to both technical and non-technical audiences.

Data Analyst Intern **Jan 2022 - Apr 2022**
Deloitte

- Implemented **T-SQL stored procedures** in SSMS to automate data transformation and aggregation, improving pipeline efficiency by 45% and reducing database query load by 30%.
- Engineered a **Python**-based anomaly detection system using **Pandas** and **NumPy** to identify inconsistencies in financial datasets, reducing reporting errors by 35% and supporting audit compliance.

TECHNICAL SKILLS

Programming & Databases: Python, **SQL (T-SQL, MySQL, PostgreSQL)**, PySpark, R, MATLAB, NoSQL (MongoDB).
ETL & Big Data: SQL Server (SSMS), SSIS, Airflow, Snowflake, Apache **Spark**, Azure Data Factory, Kafka, BigQuery, Rest APIs.
Visualization & Reporting: **Microsoft Power BI, DAX, Tableau**, Looker Studio, Alteryx, Excel, SSRS.
Cloud & DevOps: Azure (Synapse Analytics), GCP, Git, CI/CD pipelines (Jenkins), SDLC (Agile).
Data Science / ML: EDA, Predictive Modeling, Time Series Analysis (ARIMA), A/B Testing, Scikit-learn, TensorFlow, NLP
Certifications: Power BI Data Analyst Associate (PL-300), SQL Associate by Data Camp.

PROJECTS

Kaggle Ecosystem Analytics (Business Intelligence) - [Python, Azure Databricks, Azure Synapse, Power BI] **Feb 2024 - Apr 2024**

- Architected a scalable data analytics platform, ingesting 3 GB raw Kaggle data into Azure Synapse with Databricks as an intermediate layer for data cleaning and transformation.
- Designed dynamic Power BI reports using Direct Query and Import modes from Synapse, delivering real-time user behavior insights and boosting trend analysis accuracy by 70%.

Intelligent Chatbot Q/A Recommendation System - [Python, BERT LLM, NLP, Machine Learning, Data Mining] **Oct 2023 - Dec 2023**

- Developed an intelligent Q&A Chatbot that suggested relevant questions based on user input, improving user engagement by 40%.
- Applied TF-IDF and BERT for improved query vectorization, enhancing NLP-based chatbot accuracy to 90% while handling 1,000+ diverse academic queries.
- Recognized as “Best Project Presentation” for innovative use of machine learning in enhancing IU academic support services.